

Shortest Distance of Lines

Art. 29. Parametric equations of the line

Let the direction cosines of the line L be l, m, n and let $P_1(x_1, y_1, z_1)$ be a point on the line. Then L is the locus of P (x, y, z) moving so that

$$x - x_1 = r \cos \alpha = rl$$

$$y - y_1 = r \cos \beta = rm$$

$$z - z_1 = r \cos \gamma = rn$$

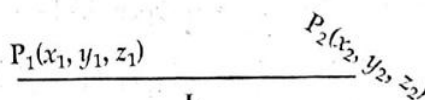


Fig. 15

$$\therefore x = x_1 + rl, y = y_1 + rm \text{ and } z = z_1 + rn \quad \dots \quad (43)$$

where the parameter represents the variable length P_1P .

If a, b, c , be direction ratios of L these equations can be written as $x = x_1 + at, y = y_1 + bt, z = z_1 + ct$ (44)

Here t does not represent the distance of the variable length P_1P .

The equation represents the co-ordinates of any point on the line L by giving different values to ' t '.

Art. 30. (a) Symmetrical Form : To find the equation of a straight line through a given point (x_1, y_1, z_1) in a given direction.

Let the given point be $P_1(x_1, y_1, z_1)$ and let the given direction be defined by the direction cosines l, m, n .

Let $P(x, y, z)$ be a point on the line and let the measure of P_1P be r .

Then projecting P_1P on the co-ordinate axes, we have

$$x - x_1 = lr, y - y_1 = mr, z - z_1 = nr$$

So that for all points (x, y, z) on the given line,

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r$$

Thus $\boxed{\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}}$

is the symmetrical form of the equation of a straight line

Art. 32. Line through two given points.

If P, Q are $(x_1, y_1, z_1), (x_2, y_2, z_2)$ the direction ratios of PQ are proportional to $x_2 - x_1, y_2 - y_1, z_2 - z_1$ and therefore the equations to PQ are $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1} \dots \dots (45)$

(b) To find the shortest distance between two given lines whose equations are given, also obtain the equations of the shortest distance.

Let ABL and CDM be the two given lines whose equations are

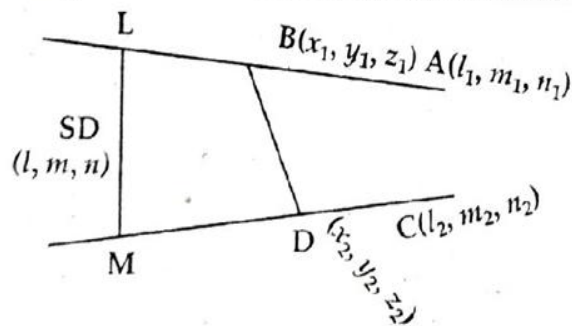
$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} \quad \dots \quad (1)$$

$$\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2} \quad \dots \quad (2)$$

Where the points B and D are $(x_1, y_1, z_1), (x_2, y_2, z_2)$

Let LM be the line which

is perpendicular to both the given lines and let its direction cosines be l, m, n



The shortest distance (S. D) i.e. LM is the projection of BD on line LM whose direction cosines are l, m, n .

$$S.D = l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

Note : If the given lines are coplanar the S. D. is zero ;

Therefore,
$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0 \quad \dots$$

The equation of the plane containing the line (1) and the line of shortest distance whose direction cosines are l, m, n ,

$$\text{is } \begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ l_1 & m_1 & n_1 \\ l & m & n \end{vmatrix} = 0 \quad \dots \quad (5) \dots \dots (63)$$

Similarly the equation of the plane containing the line (2) and the shortest distance is

$$\begin{vmatrix} x-x_2 & y-y_2 & z-z_2 \\ l_2 & m_2 & n_2 \\ l & m & n \end{vmatrix} = 0 \quad \dots \quad (6) \dots \dots (64)$$

These two planes (2) and (2) taken together represent the equation of the line of shortest distance

Ex. 14. Find the magnitude and the equations of the line of shortest distance between the lines $\frac{x-8}{3} = \frac{y+9}{-16} = \frac{z-10}{7} \dots \dots (1); \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$

Let l, m, n be the direction cosines of the lines L, M of shortest distance. As it is perpendicular two lines we have $3l - 16m + 7n = 0$ and $3l + 8m - 5n = 0$

$$\therefore 1/24 = m/36 = n/72 \text{ or } l/2 = m/3 = n/6 = 1/7$$

$$\text{Hence } l = 2/7, m = 3/7, n = 6/7$$

The magnitude of the shortest distance is the projection of the joint of the pts. $(8, -9, 10)$, and $(15, 29, 5)$ on

$$\text{ML and is } (15-8)2/7 + (29+9)3/7 + (5-10)6/7 = 10 \quad \text{or } S.D. = 10$$

Again the equation of the plane containing the first of the two given lines and the line of shortest distance is

$$\begin{vmatrix} x-8 & y+9 & z-10 \\ 3 & -16 & 7 \\ 2 & 3 & 6 \end{vmatrix} = 0 \quad \text{Or, } 117x + 4y - 41z - 490 = 0$$

Also the equation of the plane containing the second line and the S. D. is

$$\begin{vmatrix} x-15 & y-29 & z-5 \\ 3 & 8 & -5 \\ 2 & 3 & 6 \end{vmatrix} = 0 \quad \text{Or, } 9x - 4y - z - 14 = 0$$

The equation of the shortest distance is

$$117x + 4y - 41z - 490 = 0 = 9x - 4y - z - 14$$

Exercise

45. Find the shortest distance between the lines.

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}; \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5} \text{ State whether the lines are coplanar or not}$$

Ans. S. D = 0 [R. U. 1979, 81, C. H. 1977) D. H. 1980]

46. Find the equation of the line of S. D. and length of the S. D. between lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-4}{4}; \frac{x-2}{2} = \frac{y-4}{4} = \frac{z-5}{5}$$

Ans. $2/\sqrt{6}$, $11x + 2y - 7z + 13 = 0 = 7x + y - 5z + 7$.

47. Show that S. D. between the lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ is $\frac{1}{3}\sqrt{3}$

48. Find the magnitude and direction cosines of the S. D between the lines.

$$\frac{x+7}{-8} = \frac{y-5}{3} = \frac{z-4}{1} \text{ and } \frac{x+4}{4} = \frac{y}{3} = \frac{z-19}{-2}$$

Ans. 13, (3, 4, 12)